Software Requirements Specification for Software Engineering: subtitle describing software

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Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

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17.1 Legal Requirements

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17.2 Standards Compliance Requirements

Insert your content here.

18 Traceability from Requirements to Use Cases

This section maps each of the requirements to one or more of the use cases listed and defined above in previous sections.

Table 1: Traceability from Requirements to Use Cases

Requirement	Use Case(s)
FR1	PUC3, PUC10
FR2	PUC1
FR3	PUC3
FR4	PUC10
FR5	PUC8
FR6	PUC10
FR7	PUC5
FR8	PUC7
FR9	PUC9
FR10	PUC4, PUC8
FR11	PUC11
FR12	PUC11
FR13	PUC11, PUC12
FR14	PUC6
FR15	PUC7
FR16	PUC7
FR17	PUC8, PUC10
FR18	PUC10
FR19	PUC9
FR20	PUC10
FR21	PUC13
FR22	PUC10, PUC13
FR23	PUC1, PUC13
FR24	PUC13
FR25	PUC8, PUC10
LFR-AP1	All PUCs
LFR-ST1	PUC4, PUC6, PUC8
UHR-EOU1	All PUCs
UHR-EOU2	PUC6
UHR-LRN1	PUC5, PUC8
UHR-UPL1	PUC6
UHR-AR1	PUC5, PUC6, PUC7
PR-SL1	PUC5

Continued on next page

Table 1 – continued from previous page

Requirement	Use Case(s)
PR-SL2	PUC6
PR-SCR1	PUC6, PUC7
PR-PAR1	PUC11
PR-PAR2	PUC11, PUC12
PR-RFT1	PUC5
PR-RFT2	PUC5
PR-RFT3	All PUCs
PR-CR1	All PUCs
PR-SER1	PUC11, PUC12
OER-EP1	PUC5, PUC11
OER-WE1	All PUCs
OER-WE2	All PUCs
OER-PR1	PUC1, PUC3, PUC5, PUC6, PUC8, PUC10
OER-RL1	PUC5, PUC11, PUC12
MS-MNT1	PUC13
MS-AD1	PUC5, PUC8
SR-AR1	PUC1
SR-AR2	PUC3
SR-AR3	PUC2
SR-IR1	PUC13
SR-PR1	PUC1, PUC13
SR-PR2	PUC13

19 Open Issues

This section discusses the open issues in this project. In detail, we will highlight the primary uncertainties and open questions that are present throughout the process of developing our physiotherapy app. Ultimately, these issues represent the challenges we may face throughout the development of our app. Addressing these issues will aid us in ensuring the app is efficient, accurate and reliable. Listed below are some potential open issues in this project we have come across, as well as how we plan to address these issues:

- **Exercise Detection Accuracy:** Ensuring the app correctly analyzes and identifies the movements the patient undergoes is always going to be a constant challenge. Inaccuracies in the detection could result in false feedback which may prohibit recovery time. To address this, we will be undergoing further refinement of our detection methods and testing with a diverse sample size of individuals to aid in the accuracy of our movement detection to provide the most accurate feedback.

- **Data Privacy:** The app will collect and store sensitive client health information which is something that needs to be securely stored. We will be making sure information is securely stored, transmission is encrypted and there are strict access controls in place to have that trust exist with our clients.

- **Maintaining Patient Engagement:** Maintaining patient engagement is something that is going to be a struggle, as some patients may view this as extra work on their end. Our goal is to hopefully have this app be viewed as a convenient aid that reduces the need for frequent in person physiotherapy visits, making recovery more so accessible and less time-consuming for the clients.

20 Off-the-Shelf Solutions

20.1 Ready-Made Products

Insert your content here.

20.2 Reusable Components

Insert your content here.

20.3 Products That Can Be Copied

Insert your content here.

21 New Problems

21.1 Effects on the Current Environment

1. **Increased Battery Usage of Smart Device:** The app's use of motion tracking and video processing could result in an increase of battery usage for a user's smart device. This could also limit usability for individuals who have older devices.

2. **Increased Storage Impact:** Since the app does involve uploading recording videos for analysis, this may result in an increase in storage used on the smart device. Individuals using the app who may have limited storage may experience a higher amount of data being used. This ultimately means that we need to ensure that due to the increase in storage usage, the quality is not lost, as we do not want that to affect the precision of our analysis in any way, shape or form.

3. **Requirement of stable network access:** The app does rely on a stable network connection to upload exercise videos for analysis. Areas with weaker connections may lead to individuals dealing with more delays and thus, limiting their overall usability.

21.2 Effects on the Installed Systems

1. It may be hard for existing clinic or hospital systems to easily support integration with our app. There may be incompatibilities between the varying software platforms used, so this may slow down the process of integration or require more effort and money to aid in integrating it more smoothly.

2. Different systems may store data differently so this is something we also have to be aware of as it could create inconsistencies when syncing the medical records.

3. When integrating the app to healthcare networks, policies and authentication layers should be carefully managed because if proper security is not in place, unauthorized users may be able to access patient data, which is extremely unsafe.

21.3 Potential User Problems

1. **Learning Curve:** Some users may not be the most familiar with using technology and may find it a bit more difficult to use this app which ultimately may lead to a decrease in user satisfaction and engagement.

2. **Privacy Issues and Comfortability with Video Recordings:** Some users may not be comfortable with the process of recording themselves and uploading their videos to the platform, which could lead to a decrease in engagement.

3. **Preference of in-person physiotherapy guidance:** Some users may prefer in person appointments with their physiotherapists to receive direct feedback, which may lead to a decrease in user engagement overall.

4. **Limitations of Recording Videos:** Some users may find it difficult as a result of their limited mobility to be able to record and upload the videos with precision by themselves without any additional support. This may lead to inaccurate computations and results.

5. **Misinterpretation of Feedback:** Users may not understand how to correctly translate or interpret the feedback given by the app, which could ultimately lead to a decrease in the effectiveness and overall productivity of the app.

21.4 Limitations in the Anticipated Implementation Environment That May Inhibit the New Product

1. **Limited access to technology:** Some users may be limited in terms of not owning smart devices with a functional camera or not having enough storage space to record and upload videos. This will ultimately limit the app's overall accessibility.

2. **Connectivity Issues:** Areas with weaker connections lead to individuals being more prone to experiencing delays and deal with uploads failing, which ultimately decreases the app's overall reliability and effectiveness.

21.5 Follow-Up Problems

1. **Regular Updates:** The app will consist of regular updates to make sure it is compatible with the latest operating systems, all the changes in the newest programming languages as well as ensure it abides by all of the standard in place.

2. **Data Management and Cleanup:** Large amounts of data will be stored, resulting in a lot of storage being used up, as mentioned previously. Potentially introducing some sort of periodic cleanup may need to be implemented to address this.

3. **On-going support may be required:** Users may require technical or instructional support when they are faced with issues while using the app specifically when recording, uploading or even trying to interpret the results. Potentially providing client support or a FAQ section may be necessary to address this.

22 Tasks

22.1 Project Planning

Insert your content here.

22.2 Planning of the Development Phases

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26 Waiting Room

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27 Ideas for Solution

In this section , we will highlight the potential technical approaches we may be using to implement our physiotherapy app. The ideas for solution can be seen listed below:

- **Pose Detection:** We plan on using the OpenCV library to select and analyze specific frames of the video uploaded to pick and identify specific key knee joint points using pose estimation techniques. In terms of the calculations, we can use Python's built-in math module, which we can use to calculate the specific calculations necessary.

- **Feedback Charts:** To showcase graphs and progress summary charts, we can use Python's built-in matplotlib library to generate them for us. These can be later integrated to allow patients and physiotherapists the option to interpret the graphs and overall progress from the visuals provided.

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.
5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.
6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?